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Power transmitter

Abstract

A power transmitter includes: a first coil that wirelessly transmits the electric power to a second coil of a power receiver; a converter including a DC/AC converter that converts DC electric power into AC electric power and supplies the AC electric power to the first coil; and a controller that brings a power value of the DC electric power or the AC electric power close to a power command value by frequency control of changing a frequency of the AC electric power. The controller performs, when the frequency at which the power value reaches the power command value is included in a used frequency band used by another device, frequency change process of changing the frequency so that the frequency becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value matches the power command value.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a power transmitter.

BACKGROUND ART

(2) A wireless power transfer system that wirelessly transmits electric power is known. The wireless power transfer system includes a power transmitter including a power transmission coil and a power receiver including a power reception coil, and realizes power transmission in a wireless manner by utilizing electromagnetic induction or magnetic resonance between these coils. When a frequency used for power transmission (switching frequency of an inverter) interferes with a frequency used by another device, noise or the like may affect the other device due to power transmission.

(3) In the wireless power transfer device described in Patent Literature 1, when an inverter frequency corresponding to an output command value (power command value) overlaps with a used frequency band used by another device, the inverter frequency is corrected so as not to overlap with the used frequency band, and a power factor correction (PFC) output is adjusted so as to obtain an output power value corresponding to the power command value.

CITATION LIST

Patent Literature

(4) Patent Literature 1: WO 2014/118972 A

SUMMARY OF INVENTION

Technical Problem

(5) In the wireless power transfer device described in Patent Literature 1, after the inverter frequency is caused to match a frequency corresponding to the power command value, if the inverter frequency overlaps with the used frequency band, the inverter frequency is corrected so as not to overlap with the used frequency band. Then, the PFC output is increased by the amount of decrease in the output electric power due to the change in the inverter frequency, and the output power value is caused to match the power command value. Therefore, it may take a long time for the output power value to match the power command value.

(6) The present disclosure describes a power transmitter capable of reducing the time required for the power value of target electric power to match a power command value while suppressing interference with other devices.

Solution to Problem

(7) A power transmitter according to one aspect of the present disclosure is a device for supplying electric power to a power receiver connected to a load. The power transmitter includes: a first coil that wirelessly transmits the electric power to a second coil of the power receiver; a converter including a DC/AC converter that converts DC electric power into AC electric power and supplies the AC electric power to the first coil; and a controller that brings a power value of target electric power close to a power command value by frequency control of changing a frequency of the AC electric power. The target electric power is the DC electric power or the AC electric power. The controller performs, when the frequency at which the power value reaches the power command value is included in a used frequency band used by another device, frequency change process of changing the frequency so that the frequency becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value matches the power command value.

Effects of Invention

(8) According to the present disclosure, it is possible to shorten the time required for the power value of the target electric power to match the power command value while suppressing interference with other devices.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram showing an application example of a wireless power transfer system including a power transmitter according to an embodiment.
- (2) FIG. 2 is a circuit block diagram of the wireless power transfer system of FIG. 1.
- (3) FIG. 3 is a diagram showing a frequency characteristic of load electric power.
- (4) FIG. 4 is a diagram showing an example of a circuit configuration of a DC/AC converter.
- (5) FIG. 5 is a flowchart showing a series of processes of power control performed by the first controller shown in FIG. 2.
- (6) FIG. 6 is a flowchart showing an example of the frequency change process of FIG. 5 in detail.
- (7) FIG. 7 is a diagram for describing the power control of FIG. 5.
- (8) FIG. 8 is a flowchart showing another example of the frequency change process of FIG. 5 in detail.
- (9) FIG. 9 is a diagram for describing power control including the frequency change process of FIG. 8.
- (10) FIG. 10 is a circuit block diagram of a wireless power transfer system including a power transmitter according to a modification.

DESCRIPTION OF EMBODIMENTS

[1] Outline of Embodiment

(11) A power transmitter according to one aspect of the present disclosure is a device for supplying electric power to a power receiver connected to a load. The power transmitter includes: a first coil that wirelessly transmits the electric power to a second coil of the power receiver; a converter including a DC/AC converter that converts DC electric power into AC electric power and supplies the AC electric power to the first coil; and a controller that brings a power value of target electric power close to a power command value by frequency control of changing a frequency of the AC electric power. The target electric power is the DC electric power or the AC electric power. The controller performs, when the frequency at which the power value reaches the power command value is included in a used frequency band used by another device, frequency change process of changing the frequency so that the frequency becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value matches the power command value.

(12) In the power transmitter, the frequency change process is performed in a case where the frequency of the AC electric power when the power value of the target electric power reaches the power command value is included in the used frequency band used by another device. In the frequency change process, the frequency of the AC electric power is changed so that the frequency of the AC electric power becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value of the target electric power matches the power command value. In this way, once the power value of the target electric power reaches the power command value, the state in which the power value of the target electric power matches the power command value is maintained. Therefore, even if the frequency of the AC electric power is changed after the power value of the target electric power reaches the power command value, the power value of the target electric power matches the power command value. As a result, it is possible to shorten the time required for the power value of the target electric power to match the power command value while suppressing interference with other devices.

(13) The controller may perform the frequency change process by performing at least one of voltage control of changing a voltage of the DC electric power, phase shift control of the DC/AC converter, and impedance control of controlling an impedance between the DC/AC converter and

the first coil together with the frequency control. When the voltage of the DC electric power, the phase shift amount of the DC/AC converter, or the impedance between the DC/AC converter and the first coil is changed, the frequency characteristic of the target electric power is changed. Therefore, the frequency of the AC electric power is changed in order to maintain a state in which the power value of the target electric power matches the power command value. This makes it possible to set the frequency of the AC electric power to a frequency different from the used frequency band.

(14) The controller may perform the frequency change process by performing the voltage control together with the frequency control. When the voltage of the DC electric power is changed, the frequency characteristic of the target electric power is changed. Therefore, the frequency of the AC electric power is changed in order to maintain a state in which the power value of the target electric power matches the power command value. This makes it possible to set the frequency of the AC electric power to a frequency different from the used frequency band.

(15) The controller may perform the voltage control by increasing the voltage. In this case, since the voltage of the DC electric power is adjusted in order from a lower voltage, it is possible to lower the voltage of the DC electric power when the frequency of the AC electric power is changed to a frequency different from the used frequency band. As a result, the power transfer efficiency between the first coil and the second coil can be improved.

(16) The controller may perform the frequency change process by performing the phase shift control together with the frequency control. When the phase shift amount of the DC/AC converter is changed, the frequency characteristic of the target electric power is changed. Therefore, the frequency of the AC electric power is changed in order to maintain a state in which the power value of the target electric power matches the power command value. This makes it possible to set the frequency of the AC electric power to a frequency different from the used frequency band.

(17) The controller may perform the frequency change process by performing the impedance control together with the frequency control. When the impedance between the DC/AC converter and the first coil is changed, the frequency characteristic of the target electric power is changed. Therefore, the frequency of the AC electric power is changed in order to maintain a state in which the power value of the target electric power matches the power command value. This makes it possible to set the frequency of the AC electric power to a frequency different from the used frequency band.

(18) The controller may control the converter to keep the voltage constant. In this case, since the voltage of the DC electric power is constant, the frequency change process can be performed without increasing the withstand voltage of the DC/AC converter. As a result, it is possible to suppress interference with other devices without increasing the size of the DC/AC converter.

(19) The controller may select one of the voltage control and the phase shift control in accordance with the frequency when the power value reaches the power command value, and may perform the frequency change process by performing the selected control together with the frequency control. In this case, it is possible to shorten the time until the frequency of the AC electric power is set to a frequency different from the used frequency band. Since the adjustment range of the frequency of the AC electric power can be widened by using the voltage control and the phase shift control, it is possible to more reliably set the frequency of the AC electric power to a frequency different from the used frequency band.

(20) The controller may select one of the voltage control and the impedance control in accordance with the frequency when the power value reaches the power command value, and may perform the frequency change process by performing the selected control together with the frequency control. In this case, it is possible to shorten the time until the frequency of the AC electric power is set to a frequency different from the used frequency band. Since the adjustment range of the frequency of the AC electric power can be widened by using the voltage control and the impedance control, it is possible to more reliably set the frequency of the AC electric power to a frequency different from

the used frequency band.

(21) The controller may lower the power command value when the controller cannot change the frequency to any frequency different from the used frequency band. In this case, the frequency of the AC electric power can be changed to a frequency different from the used frequency band by lowering the power command value. Therefore, it is possible to more reliably suppress interference with other devices.

[2] Examples of Embodiments

(22) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the description of the drawings, the same elements are denoted by the same reference numerals and redundant explanation is omitted.

(23) FIG. 1 is a diagram showing an application example of a wireless power transfer system including a power transmitter according to an embodiment. As shown in FIG. 1, a wireless power transfer system 1 includes a power transmitter 2 and a power receiver 3, and is a system for supplying electric power from the power transmitter 2 to the power receiver 3. The power transmitter 2 and the power receiver 3 are separated from each other, for example, in a vertical direction. The power transmitter 2 is installed in a parking lot or the like, for example. The power receiver 3 is mounted on an electric vehicle EV, for example. The wireless power transfer system 1 is configured to supply electric power to the electric vehicle EV arriving at the parking lot or the like, using inter-coil magnetic coupling such as a magnetic field resonance method and an electromagnetic induction method. The wireless power transfer method is not limited to a method utilizing the magnetic coupling, but may be an electric field resonance method, for example.

(24) The power transmitter 2 is a device that supplies electric power for wireless power transfer. The power transmitter 2 generates desired AC electric power from electric power supplied by a power source PS (see FIG. 2), and transmits the generated AC electric power to the power receiver 3. The power transmitter 2 is installed on a road surface R such as a parking lot. The power transmitter 2 includes a first coil device 4 (power transmission coil device) provided so as to protrude upward from the road surface R such as the parking lot. The first coil device 4 includes a first coil 21 (see FIG. 2), and has a flat frustum shape or a rectangular parallelepiped shape, for example. The power transmitter 2 generates the desired AC electric power from the power source PS. The generated AC electric power is transmitted to the first coil device 4, so that the first coil device 4 generates a magnetic flux.

(25) The power receiver 3 is a device that receives electric power from the power transmitter 2 and supplies electric power to a load L (see FIG. 2). The power receiver 3 is mounted on the electric vehicle EV, for example. The power receiver 3 includes, for example, a second coil device 5 (power receiving coil device) attached to a bottom surface of a vehicle body (chassis or the like) of the electric vehicle EV. The second coil device 5 includes a second coil 31 (see FIG. 2), and faces the first coil device 4 while being spaced apart from each other in a vertical direction when electric power is supplied. The second coil device 5 has, for example, a flat frustum shape or a rectangular parallelepiped shape. The magnetic flux generated in the first coil device 4 interlinks with the second coil device 5, so that the second coil device 5 generates an induced electric current. Thereby, the second coil device 5 receives electric power from the first coil device 4 in a wireless manner. The electric power received by the second coil device 5 is supplied to the load L.

(26) With reference to FIG. 2, a circuit configuration of the wireless power transfer system 1 will be described in detail. FIG. 2 is a circuit block diagram of the wireless power transfer system of FIG. 1. As shown in FIG. 2, the wireless power transfer system 1 is a system for receiving AC electric power P_{ac1} from the power source PS and supplying load electric power P_{out} to the load L. The power source PS may be an AC electric power source such as a commercial power source, and supplies the AC electric power P_{ac1} to the power transmitter 2. A frequency of the AC electric power P_{ac1} is, for example, 50 Hz or 60 Hz. The load L may be a DC load such as a battery or an AC load such as an electric motor.

(27) The power transmitter **2** is supplied with the AC electric power P_{ac1} from the power source PS. The power transmitter **2** includes the first coil **21**, a first converter **22** (converter), a first detector **23**, a first communication device **24**, a first controller **25** (controller), and a used frequency detector **28**.

(28) The first converter **22** is a circuit that converts the AC electric power P_{ac1} supplied from the power source PS into desired AC electric power P_{ac2} and supplies the AC electric power P_{ac2} to the first coil **21**. The first converter **22** can change a magnitude (power value) of the DC electric power P_{dc} and a magnitude (power value) of the AC electric power P_{ac2} by, for example, a frequency control, a phase shift control, and a voltage control of the DC electric power P_{dc} , which will be described later. The first converter **22** includes a power converter **26** and a DC/AC converter **27**.

(29) The power converter **26** is an AC/DC converter that converts the AC electric power P_{ac1} supplied from the power source PS into the DC electric power P_{dc} . The power converter **26** is, for example, a rectifier circuit. The rectifier circuit may be constituted by a rectifying element such as a diode or may be constituted by a switching element such as a transistor. The power converter **26** may further include a power factor correction (PFC) function and a step-up/step-down function. The first converter **22** may further include a DC/DC converter that is provided at an output of the power converter **26**. The power converter **26** is controlled by the first controller **25** so as to change a magnitude of a voltage V_{dc} of the DC electric power P_{dc} . The power converter **26** changes the magnitude of the voltage V_{dc} of the DC electric power P_{dc} by pulse width modulation, for example. The power converter **26** supplies the DC electric power P_{dc} to the DC/AC converter **27**.

(30) The DC/AC converter **27** converts the DC electric power P_{dc} supplied from the power converter **26** into the AC electric power P_{ac2} . A frequency of the AC electric power P_{ac2} is, for example, 81.38 kHz to 90 kHz. The DC/AC converter **27** includes, for example, an inverter circuit. The first converter **22** may further include a transformer that is provided at the output of the DC/AC converter **27**. The DC/AC converter **27** is controlled by the first controller **25** so as to change the magnitudes of the DC electric power P_{dc} and the AC electric power P_{ac2} . The DC/AC converter **27** supplies the AC electric power P_{ac2} to the first coil **21**.

(31) The first coil **21** is a coil for wirelessly transmitting electric power to (the second coil **31** of) the power receiver **3**. The first coil **21** generates a magnetic flux by supplying the AC electric power P_{ac2} from the first converter **22** to the first coil **21**. A capacitor and an inductor (for example, a reactor) may be connected between the first coil **21** and the first converter **22**.

(32) The first detector **23** includes a circuit for acquiring a measurement value relating to the DC electric power P_{dc} . The circuit for acquiring the measurement value relating to the DC electric power P_{dc} is, for example, a voltage sensor, a current sensor, or a combination thereof. The first detector **23** measures the DC electric power P_{dc} , the voltage V_{dc} of the DC electric power P_{dc} , or an electric current I_{dc} of the DC electric power P_{dc} . The first detector **23** measures the AC electric power P_{ac2} , a voltage V_{ac2} of the AC electric power P_{ac2} , and an electric current I_{ac2} of the AC electric power P_{ac2} . The first detector **23** outputs the acquired measurement value to the first controller **25**.

(33) The first communication device **24** is a circuit for wirelessly communicating with a second communication device **34** of the power receiver **3** to be described later. The first communication device **24** includes, for example, an antenna for a communication system using radio waves, or a light emitting element and a light receiving element for a communication system using an optical signal. The first communication device **24** outputs the information received from the power receiver **3** to the first controller **25**.

(34) The first controller **25** is a processing device such as a central processing unit (CPU) and a digital signal processor (DSP). The first controller **25** may include a read only memory (ROM), a random access memory (RAM), an interface circuit connected to each part of the power transmitter **2**, and the like.

(35) The first controller **25** performs constant power control to bring the power value of the target electric power close to the first power command value (power command value). As the target electric power, the DC electric power P_{dc} or the AC electric power P_{ac2} may be used. The first controller **25** calculates the first power measurement value based on the measurement value of the electric current I_{dc} detected by the first detector **23**. When the target electric power is the DC electric power P_{dc} , the first power measurement value is a measurement value including a loss of the DC/AC converter **27**, a loss of the first coil **21**, and the like and the AC electric power P_{ac2} supplied from the DC/AC converter **27** to the first coil **21**. The first controller **25** calculates the first power command value, which is a target value of the DC electric power P_{dc} , based on a second power command value received from the power receiver **3** via the first communication device **24**. As the constant power control, the first controller **25** performs the power control that controls the first converter **22** based on the first power measurement value and the first power command value so that the first power measurement value (DC electric power P_{dc}) approaches the first power command value.

(36) When the target electric power is the AC electric power P_{ac2} , the first power command value is a target value of the AC electric power P_{ac2} and is set in accordance with the AC electric power P_{ac2} . That is, as the constant power control, the first controller **25** controls the first converter **22** so that the AC electric power P_{ac2} approaches the first power command value, which is the target value of the AC electric power P_{ac2} . Hereinafter, a case where the target electric power is the DC electric power P_{dc} will be described.

(37) The first controller **25** may perform a command value correction control for correcting the first power command value. As the command value correction control, the first controller **25** performs the power control that controls the first converter **22** based on the second power measurement value (described later) and the second power command value (described later) received from the power receiver **3** via the first communication device **24** so that the second power measurement value (load electric power P_{out}) approaches the second power command value. Specifically, the first controller **25** corrects the first power command value so that the second power measurement value approaches the second power command value.

(38) As the power control, the first controller **25** controls the magnitude (power value) of the DC electric power P_{dc} and the magnitude (power value) of the AC electric power P_{ac2} by controlling the first converter **22** and controls the magnitude of the load electric power P_{out} supplied to the load **L**. The power control is performed using at least one of the frequency control, the phase shift control, and the voltage control of the DC electric power P_{dc} . In each control, a power control parameter for controlling the magnitudes of the DC electric power P_{dc} and the AC electric power P_{ac2} is changed.

(39) Here, the frequency control will be described with reference to FIG. 3. FIG. 3 is a diagram showing a frequency characteristic of load electric power. A horizontal axis of a graph in FIG. 3 indicates the driving frequency f , and a vertical axis thereof indicates (the magnitude of) the load electric power P_{out} . The driving frequency f is the frequency of the AC electric power P_{ac2} . The frequency of the AC electric power P_{ac2} refers to a frequency of the AC electric current or the AC voltage that is output from the first converter **22**. Hereinafter, the frequency of the AC electric power P_{ac2} may be referred to as “driving frequency f ”. As shown in FIG. 3, the magnitudes of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} are changed with the driving frequency f . As the driving frequency f , for example, 81.38 kHz to 90 kHz are available. As the driving frequency f is changed, an impedance of a reactance element such as the coil and the capacitor is changed, whereby the magnitudes of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} are changed. Hereinafter, in present embodiment, as the driving frequency f increases, the magnitudes of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} decrease. The first controller **25** performs the frequency control that changes the magnitudes (power values) of the DC electric power P_{dc} , the

AC electric power P_{ac2} , and the load electric power P_{out} by changing the driving frequency f . In the frequency control, the above-described power control parameter is the driving frequency f of the DC/AC converter **27** (inverter circuit).

(40) For example, it is assumed that the driving frequency f is initially a frequency f_b . The load electric power P_{out} at this time is an electric power P_b . Here, for example, the driving frequency f is reduced from the frequency f_b to a frequency f_a . Then, the load electric power P_{out} becomes an electric power P_a corresponding to the driving frequency $f=f_a$. Therefore, the load electric power P_{out} increases from the electric power P_b to the electric power P_a . On the other hand, the driving frequency f is increased from the frequency f_b to a frequency f_c . Then, the load electric power P_{out} becomes an electric power P_c corresponding to the driving frequency $f=f_c$. Therefore, the load electric power P_{out} decreases from the electric power P_b to the electric power P_c .

(41) The first controller **25** brings the load electric power P_{out} closer to a desirable electric power by changing the driving frequency f as described above. In the control that actually changes (increases and decreases) the driving frequency f , the driving frequency f may be changed by steps. The size of one step for changing the driving frequency f is not particularly limited, and may be, for example, about several hertz to several tens of hertz or several tens of hertz to several hundreds of hertz. The steps may be determined, for example, by a clock resolution of the CPU, which is the first controller **25**.

(42) A specific procedure of the frequency control is not limited. For example, in a case where the DC/AC converter **27** is an inverter circuit, the first controller **25** adjusts the switching frequency of each switching element using the driving signal that is supplied to each of the switching elements included in the inverter circuit to change the driving frequency f . As the switching element, for example, a field effect transistor (FET) and an insulated gate bipolar transistor (IGBT) are used. In this case, the driving signal is applied to a gate of the switching element.

(43) Next, the phase shift control will be described. The first controller **25** performs the phase shift control that changes the magnitudes (power values) of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} by changing the phase shift amount of the DC/AC converter **27** (inverter circuit). For example, in a case where the DC/AC converter **27** is an inverter circuit as shown in FIG. 4, the first controller **25** adjusts the supply times of the driving signals S_a to S_d to the switching elements SW_a to SW_d included in the inverter circuit to adjust the time in which each of the switching elements SW_a to SW_d is turned on.

(44) When a driving time of the switching element SW_a and a driving time of the switching element SW_d are the same and a driving time of the switching element SW_b and a driving time of the switching element SW_c are the same, an energization period (ON period) of the inverter circuit is longest. The ON period of the inverter circuit becomes shorter as the driving time of the switching element SW_a and the driving time of the switching element SW_d are shifted (the driving time of the switching element SW_b and the driving time of the switching element SW_c are shifted). As the ON period of the inverter circuit becomes shorter, the DC electric power P_{dc} and the AC electric power P_{ac2} become smaller. The above-described power control parameter in the phase shift control is a phase shift amount. The phase shift amount is a shift amount between the driving time of the switching element SW_a and the driving time of the switching element SW_d (or a shift amount between the driving time of the switching element SW_b and the driving time of the switching element SW_c). In other words, the above-described power control parameter in the phase shift control is the ON period of the inverter circuit.

(45) In order to realize soft switching of the inverter circuit, it is necessary that the phase of the output voltage (the voltage V_{ac2} of the AC electric power P_{ac2}) from the inverter circuit is the same as or leads the phase of the output electric current (the electric current I_{ac2} of the AC electric power P_{ac2}) (the impedance of the entire load viewed from the inverter circuit is inductive). Here, in the present embodiment, the impedance of the entire load viewed from the inverter circuit (DC/AC converter **27**) is the sum of the impedances of the first coil **21**, the second coil **31**, a

second converter **32**, and the load **L**. If the phase difference between the voltage and the electric current is kept the same, the impedance of the entire load viewed from the inverter circuit becomes capacitive due to noise, control error, and the like. When the phase of the output electric current (the electric current I_{ac2} of the AC electric power P_{ac2}) from the inverter circuit leads the phase of the output voltage (the voltage V_{ac2} of the AC electric power P_{ac2}), the impedance of the entire load viewed from the inverter circuit is capacitive. Therefore, in order to ensure safety, the phase of the voltage leads the phase of the electric current so that the phase difference does not become smaller than a predetermined value. This predetermined value is referred to as a phase margin.

(46) For example, the phase shift amount may be expressed in a percentage in which a duration of one cycle of the AC power P_{ac2} (that is, 360 degrees) is 100%. In this case, the phase shift amount is 0% in a state in which no phase shift is performed. In the phase shift control, when the phase shift amount is 0%, the DC electric power P_{dc} and the AC electric power P_{ac2} are maximized and the load electric power P_{out} is also maximized. The maximum value of the phase shift amount varies depending on the circuit characteristics of the first coil **21** (for example, characteristics of a resonance circuit including the first coil **21** and a capacitor (not shown)), but is, for example, about 50%. That is, in one aspect, the lower limit value of the phase shift amount may be set to 0%. The upper limit value of the phase shift amount may be set to 50%.

(47) Next, the voltage control of the DC electric power P_{dc} will be described. The first controller **25** performs the voltage control that changes the magnitudes (power values) of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} by changing the magnitude of the voltage V_{dc} of the DC electric power P_{dc} . The voltage V_{dc} of the DC electric power P_{dc} is changed by using, for example, the step-up/step-down function of the power converter **26** described above. For example, as the voltage V_{dc} of the DC electric power P_{dc} increases, the DC electric power P_{dc} and the AC electric power P_{ac2} also increase, and as the voltage V_{dc} of the DC electric power P_{dc} decreases, the DC electric power P_{dc} and the AC electric power P_{ac2} also decrease. Therefore, the above-described power control parameter in the voltage control of the DC electric power P_{dc} is the magnitude of the voltage V_{dc} of the DC electric power P_{dc} . The step-up/step-down function can be realized by, for example, a chopper circuit.

(48) The used frequency detector **28** is a circuit that detects a used frequency band, which is a frequency band used by another device different from the wireless power transfer system **1**. The used frequency detector **28** detects a used frequency band by automatic scanning in the same manner as automatic tuning of a radio, for example. The used frequency detector **28** may acquire positional information of the power transmitter **2** using a global positioning system (GPS) or the like, and detect, as the used frequency band, an allocated frequency band allocated to another device in an area including the position indicated by the positional information. The allocated frequency band for each area is set in advance in a memory (not shown). The used frequency detector **28** may use a harmonic component of the detected frequency band as the used frequency band. The used frequency detector **28** outputs used frequency information indicating a used frequency band to the first controller **25**. Alternatively, the user may set used frequencies in the first controller **25**. In this case, the power transmitter **2** may not include the used frequency detector **28**.

(49) The power receiver **3** includes the second coil **31**, the second converter **32**, a second detector **33**, the second communication device **34**, and a second controller **35**.

(50) The second coil **31** is a coil for receiving power wirelessly supplied from the power transmitter **2**. The magnetic flux generated by the first coil **21** interlinks with the second coil **31** to generate AC electric power P_{ac3} in the second coil **31**. The second coil **31** supplies the AC electric power P_{ac3} to the second converter **32**. A capacitor and an inductor (for example, a reactor) may be connected between the second coil **31** and the second converter **32**.

(51) The second converter **32** is a circuit that converts the AC electric power P_{ac3} supplied from the second coil **31** into the desired load electric power P_{out} for the load **L**. When the load **L** is a DC load, the second converter **32** is an AC/DC converter (rectifier circuit) that converts the AC electric

power Pac3 into the DC load electric power Pout. In this case, the second converter 32 may include a step-up/step-down function for outputting the desired load electric power Pout for the load L. The step-up/step-down function can be realized by, for example, a chopper circuit or a transformer. The second converter 32 may further include a transformer that is provided at the input of the AC/DC converter.

(52) When the load L is an AC load, the second converter 32 further includes a DC/AC converter (inverter circuit) in addition to an AC/DC converter that converts the AC electric power Pac3 into DC electric power. The DC/AC converter converts the DC electric power generated by the AC/DC converter into AC load electric power Pout. The second converter 32 may further include a transformer that is provided at the input of the AC/DC converter. When the AC electric power Pac3 supplied from the second coil 31 is the desired AC electric power for the load L, the second converter 32 can be omitted.

(53) The second detector 33 is a circuit for acquiring the measurement value relating to the load electric power Pout supplied to the load L. The second detector 33 measures a load voltage Vout, a load electric current Tout, or the load electric power Pout supplied to the load L. The second detector 33 is, for example, a voltage sensor, a current sensor, or a combination thereof. The second detector 33 outputs the acquired measurement value to the second controller 35. The load L outputs the second power command value to the second controller 35. The second power command value indicates the magnitude of the desired electric power to be supplied to the load L. For example, when the load L is a storage battery, the second power command value may be a command value of electric current, voltage, or electric power which is set in accordance with a state of charge (SOC) of the load L.

(54) The second communication device 34 is a circuit for wirelessly communicating with the first communication device 24 of the power transmitter 2. The power receiver 3 can communicate with the power transmitter 2 by the second communication device 34. The second communication device 34 includes, for example, an antenna for a communication system using radio waves, or a light emitting element and a light receiving element for a communication system using an optical signal. The second communication device 34 transmits the information received from the second controller 35 to the power transmitter 2.

(55) The second controller 35 is a processing device such as a CPU and a DSP. The second controller 35 may include a ROM, a RAM, an interface circuit connected to each unit of the power receiver 3, and the like. The second controller 35 calculates the second power measurement value based on the measurement value received from the second detector 33. The second controller 35 transmits the second power measurement value and the second power command value received from the load L to the power transmitter 2 via the second communication device 34.

(56) For example, when a storage battery of an electric vehicle is connected to the power transmitter 2 instead of the power source PS, and the power source PS is connected to the power receiver 3 instead of the load L, electric power can be transmitted from the power receiver 3 to the power transmitter 2.

(57) Next, the power control will be described in detail with reference to FIGS. 5 to 7. FIG. 5 is a flowchart showing a series of processes of power control performed by the first controller shown in FIG. 2. FIG. 6 is a flowchart showing an example of the frequency change process of FIG. 5 in detail. FIG. 7 is a diagram for describing the power control of FIG. 5. The series of processes shown in FIG. 5 is started when the first controller 25 receives the second power command value from the power receiver 3.

(58) First, the first controller 25 calculates the first power command value based on the second power command value received from the power receiver 3 (step S1). When the power receiver 3 transmits the second power measurement value to the power transmitter 2 together with the second power command value, the first controller 25 may correct the first power command value so that the second power measurement value approaches the second power command value. Here, a case

where the first power measurement value (DC electric power P_{dc}) is smaller than the first power command value will be described, but the same applies to a case where the first power measurement value is larger than the first power command value. When the first power measurement value is equal to the first power command value, the subsequent processes are not performed, and the power control ends.

(59) Subsequently, the first controller **25** sets the voltage V_{dc} (step S2). For example, when the power transmitter **2** is activated, the first controller **25** controls the power converter **26** so that the voltage V_{dc} becomes the minimum voltage in the voltage range of the voltage V_{dc} that can be output by the power converter **26**. The first controller **25** may control the power converter **26** so that the voltage V_{dc} becomes a predetermined voltage (for example, 420V). When the power transmitter **2** is operating, the first controller **25** may fix the voltage V_{dc} output by the power converter **26**.

(60) Here, a case where the voltage V_{dc} is set to the minimum voltage in the voltage range that can be output by the power converter **26** will be described as an example. At this voltage V_{dc} , the DC electric power P_{dc} has a characteristic C1 shown in FIG. 7. The characteristics C1 to C4 shown in FIG. 7 are frequency characteristics of the DC electric power P_{dc} at different voltages V_{dc} . As the voltage V_{dc} increases, the frequency characteristic of the DC electric power P_{dc} is changed in the order of the characteristic C1, the characteristic C2, the characteristic C3, and the characteristic C4. In other words, as the voltage V_{dc} increases, the driving frequency f for obtaining the same DC electric power P_{dc} increases.

(61) Subsequently, the first controller **25** performs the frequency control (step S3) to bring the first power measurement value close to the first power command value. Here, the first power measurement value (the power value of the DC electric power P_{dc}) changes along the characteristic C1 as the driving frequency f is changed. When the power transmitter **2** is activated, the driving frequency f before the frequency control is performed is set to a frequency (for example, 90 kHz) at which the DC electric power P_{dc} becomes the smallest. When the power transmitter **2** is in operation, the driving frequency f used until immediately before may be used as it is. The first controller **25** gradually brings the first power measurement value close to the first power command value, for example, by changing the driving frequency f step by step. The first controller **25** may increase the variation amount of the driving frequency f per step in a frequency band where the variation amount of the DC electric power P_{dc} with respect to the variation amount of the driving frequency f is small.

(62) As shown in FIG. 7, in the process of causing the first power measurement value to match the first power command value, the driving frequency f may be temporarily included in the used frequency band (in this example, the used frequency band FB2). However, since the driving frequency f is included in the used frequency band FB2 for a relatively short time, the influence on other devices is limited.

(63) Then, the first controller **25** determines whether or not the first power measurement value has reached (matches) the first power command value (step S4). When it is determined that the first power measurement value has not reached (does not match) the first power command value (step S4; NO), the first controller **25** continues to perform the frequency control of the step S3. On the other hand, when it is determined that the first power measurement value has reached (matches) the first power command value (step S4; YES), the first controller **25** reads the used frequency band information from the memory (not shown) and determines whether or not the frequency f_0 of the driving frequency f is included in any of the used frequency bands (step S5). The frequency f_0 is a driving frequency f when the power value of the DC electric power P_{dc} reaches the first power command value in the characteristic C1.

(64) When it is determined that the frequency f_0 is not included in any of the used frequency bands (step S5; NO), a series of processes of the power control ends. On the other hand, when it is determined in step S5 that the frequency f_0 is included in any of the used frequency bands (step S5;

YES), the first controller **25** performs a frequency change process (step **S6**) for setting the driving frequency f to be outside the used frequency band. The frequency change process is a process of changing the driving frequency f to a frequency different from the used frequency band while performing the constant power control by the frequency control.

(65) In the frequency change process of step **S6**, as shown in FIG. **6**, first, the first controller **25** performs the voltage control of the DC electric power P_{dc} while performing the constant power control by the frequency control (step **S11**). To be specific, the first controller **25** controls the power converter **26** to change the magnitude of the voltage V_{dc} . In step **S2**, when the voltage V_{dc} is set to the minimum voltage in the voltage range of the voltage V_{dc} that can be output by the power converter **26**, the first controller **25** controls the power converter **26** to increase the voltage V_{dc} . The first controller **25** gradually changes the voltage V_{dc} step by step, for example.

(66) At this time, the frequency characteristic changes according to the magnitude of the voltage V_{dc} . Therefore, when the driving frequency f is maintained at the frequency f_0 , the power value (first power measurement value) of the DC electric power P_{dc} becomes a value different from the first power command value. However, since the first controller **25** performs the constant power control using the frequency control, the first controller **25** controls the power converter **26** so as to change the magnitude of the voltage V_{dc} and maintains a state in which the first power measurement value matches the first power command value by changing the driving frequency f .

(67) Subsequently, the first controller **25** determines whether or not the driving frequency f becomes a frequency outside the used frequency band (step **S12**). When it is determined that the driving frequency f is not any frequency outside the used frequency band, that is, the driving frequency f is included in the used frequency band (step **S12**; NO), the first controller **25** determines whether or not the voltage V_{dc} can be changed (step **S13**). For example, as shown in FIG. **7**, when the voltage V_{dc} is increased from the minimum voltage in the voltage range of the voltage V_{dc} that can be output by the power converter **26**, the voltage V_{dc} can be further increased if the voltage V_{dc} does not reach the maximum voltage in the voltage range. In such a case, the first controller **25** determines that the voltage V_{dc} can be changed (step **S13**; YES), and then performs the process of step **S11** again.

(68) In the example of FIG. **7**, the frequency characteristic of the DC electric power P_{dc} is changed to the characteristic **C2** by increasing the voltage V_{dc} . However, in the characteristic **C2**, the driving frequency f when the power value of the DC electric power P_{dc} matches the first power command value is included in the used frequency band **FB1**. Therefore, the voltage V_{dc} is further increased.

(69) On the other hand, in the above-described case, if the voltage V_{dc} reaches the maximum voltage of the voltage range, the voltage V_{dc} cannot be further increased. In such a case, the first controller **25** determines that the voltage V_{dc} cannot be changed (step **S13**; NO), changes the first power command value (step **S14**), and then performs the process of step **S11** again. For example, the first controller **25** changes (lowers) the first power command value so that the driving frequency f is set to a frequency outside the used frequency band in a range in which the voltage V_{dc} can be changed.

(70) On the other hand, when it is determined in step **S12** that the driving frequency f becomes a frequency outside the used frequency band (step **S12**; YES), a series of processes of the power control ends. In the example of FIG. **7**, the frequency characteristic of the DC electric power P_{dc} is changed to the characteristic **C4** by increasing the voltage V_{dc} . In the characteristic **C4**, the driving frequency f (frequency f_1) when the power value of the DC electric power P_{dc} matches the first power command value is outside the used frequency band **FB1**.

(71) For example, in the case where the load **L** is a battery, the battery voltage (load voltage V_{out}) increases as the charging of the battery proceeds. In this case, although the frequency characteristic of the DC electric power P_{dc} changes, the driving frequency f changes because the constant power control by the frequency control is performed. At this time, whether the driving frequency f

increases or decreases is determined according to the configuration of the resonance circuit or the like. When the driving frequency f increases, the driving frequency f moves away from the used frequency band FB1. In this case, if the first power command value is decreased in step S14, the first controller 25 may increase the first power command value to the original value in a step-by-step manner. When the driving frequency f decreases, the driving frequency f approaches the used frequency band FB1. Therefore, the first controller 25 may further decrease the first power command value so that the driving frequency f is not included in the used frequency band FB1. When the driving frequency f decreases to some extent, the first power command value may be smaller than the lower limit frequency of the used frequency band FB1 even if the first power command value is the original value. Therefore, the first controller 25 may return the first power command value to the original value when the decrease amount of the driving frequency f exceeds a preset threshold value.

(72) As described above, in the power transmitter 2, the frequency change process is performed when the driving frequency f (frequency f_0) of the AC electric power P_{ac2} at the time when the power value (first power measurement value) of the DC electric power P_{dc} reaches the first power command value is included in the used frequency band used by another device. In the frequency change process, the driving frequency f is changed so that the driving frequency f becomes a frequency different from the used frequency band while performing the constant power control for maintaining a state in which the power value of the DC electric power P_{dc} matches the first power command value. As described above, once the power value of the DC electric power P_{dc} reaches the first power command value, the state in which the power value of the DC electric power P_{dc} matches the first power command value is maintained. Therefore, even if the driving frequency f is changed after the power value of the DC electric power P_{dc} reaches the first power command value, the power value of the DC electric power P_{dc} matches the first power command value. As a result, it is possible to shorten the response time from when the second power command value is received to when the power value of the DC electric power P_{dc} matches the first power command value while suppressing interference with other devices.

(73) In the above-described embodiment, the first controller 25 performs the frequency change process by performing the voltage control of the DC electric power P_{dc} together with the frequency control. When the voltage V_{dc} of the DC electric power P_{dc} is changed, the frequency characteristic of the DC electric power P_{dc} is changed. Therefore, the driving frequency f is changed in order to maintain a state in which the power value of the DC electric power P_{dc} matches the first power command value. Thus, the driving frequency f can be set to a frequency different from the used frequency band.

(74) In the above-described embodiment, the first controller 25 sets the voltage V_{dc} to the minimum voltage within the voltage range that can be output by the power converter 26. Therefore, the first controller 25 performs the voltage control of the DC electric power P_{dc} by increasing the voltage V_{dc} in the frequency change process. In this case, since the voltage V_{dc} is increased in order from a lower voltage, it is possible to lower the voltage V_{dc} when the driving frequency f is changed to a frequency different from the used frequency band. The lower the voltage V_{dc} is, the higher the power transfer efficiency between the first coil 21 and the second coil 31 is. Therefore, the power transfer efficiency between the first coil 21 and the second coil 31 can be improved. By lowering the voltage V_{dc} , the possibility of breakdown of the power transmitter 2 can be reduced.

(75) When the first controller 25 cannot change the driving frequency f to any frequency different from the used frequency band, the first controller 25 lowers the first power command value and then performs frequency change process. Thus, the driving frequency f can be changed to a frequency different from the used frequency band. Therefore, it is possible to more reliably suppress interference with other devices.

(76) When the gap between the first coil 21 and the second coil 31 varies, the load electric power P_{out} may vary. In the case where the load L is a battery, when the load voltage V_{out} varies in

accordance with the SOC of the battery, the load electric power P_{out} may also vary. On the other hand, since the first controller **25** performs the constant power control after the power value of the DC electric power P_{dc} reaches the first power command value, it is possible to follow the transient response as described above.

(77) Although the embodiments of the present disclosure have been described above, the present disclosure is not limited to the above-described embodiments. For example, the wireless power transfer system **1** may be applied not only to the electric vehicle EV but also to a moving body such as a plug-in hybrid vehicle or an underwater vehicle, or may be applied to a device other than the moving body.

(78) The first controller **25** may perform the frequency change process by performing the phase shift control together with the frequency control instead of the voltage control of the DC electric power P_{dc} . The power control of a modification will be described with reference to FIGS. **5**, **8**, and **9**. FIG. **8** is a flowchart showing another example of the frequency change process of FIG. **5** in detail. FIG. **9** is a diagram for describing power control including the frequency change process of FIG. **8**. The initial value of the phase shift amount of the DC/AC converter **27** is set to 0.

(79) In the power control of the modification, steps **S1** to **S5** are the same as steps **S1** to **S5** of the power control of the above-described embodiment. In this modification, a case where the voltage V_{dc} is set (fixed) to the minimum voltage in the voltage range that can be output by the power converter **26** will be described as an example.

(80) Subsequently, the first controller **25** performs the frequency change process (step **S6**). In the frequency change process in step **S6**, as shown in FIG. **8**, first, the first controller **25** performs the phase shift control of the DC/AC converter **27** while performing the constant power control by the frequency control (step **S21**). To be specific, the first controller **25** changes the phase shift amount by adjusting the supply time of the driving signals S_a to S_d (see FIG. **4**). Since the initial value of the phase shift amount is set to 0, the first controller **25** increases the phase shift amount. For example, the first controller **25** gradually changes the phase shift amount step by step.

(81) At this time, the frequency characteristic of the DC electric power P_{dc} changes in accordance with the phase shift amount. Therefore, when the driving frequency f is maintained at the frequency f_0 , the power value (first power measurement value) of the DC electric power P_{dc} becomes a value different from the first power command value. For example, when the phase shift amount is increased, the power value of the DC electric power P_{dc} decreases even if the DC/AC converter **27** is driven at the same driving frequency f . However, since the first controller **25** performs the constant power control using the frequency control, the first controller **25** maintains a state in which the first power measurement value matches the first power command value by changing (decreasing) the driving frequency f while changing the phase shift amount.

(82) Subsequently, the first controller **25** determines whether or not the driving frequency f becomes a frequency outside the used frequency band (step **S22**). When it is determined that the driving frequency f is not any frequency outside the used frequency band, that is, the driving frequency f is included in the used frequency band (step **S22**; NO), the first controller **25** determines whether or not the phase shift amount can be changed (increased) (step **S23**). If the phase shift amount has not reached the maximum value, the phase shift amount can be further increased. In such a case, the first controller **25** determines that the phase shift amount can be changed (step **S23**; YES), and then performs the process of step **S21** again.

(83) On the other hand, if the phase shift amount reaches the maximum value, the phase shift amount cannot be further increased. In such a case, the first controller **25** determines that the phase shift amount cannot be changed (step **S23**; NO), changes the first power command value (step **S24**), and then performs the process of step **S21** again. For example, the first controller **25** lowers the first power command value so that the driving frequency f is set to a frequency outside the used frequency band within a range in which the phase shift amount can be changed.

(84) On the other hand, when it is determined in step **S22** that the driving frequency f becomes a

frequency outside the used frequency band (step S22; YES), a series of processes of the power control ends. In the example of FIG. 9, the frequency characteristic of the DC electric power P_{dc} is changed by increasing the phase shift amount. The driving frequency f (frequency f_2) in the case where the power value of the DC electric power P_{dc} matches the first power command value in this characteristic is smaller than the driving frequency f (frequency f_0) in the case where the power value of the DC electric power P_{dc} matches the first power command value in the characteristic C1, and is outside the used frequency band FB1.

(85) For example, in the case where the load L is a battery, the battery voltage (load voltage V_{out}) increases as the charging of the battery proceeds. In this case, although the frequency characteristic of the DC electric power P_{dc} changes, the driving frequency f changes because the constant power control by the frequency control is performed. When the driving frequency f decreases, the driving frequency f moves away from the used frequency band FB1. In this case, the first controller 25 may reduce the phase shift amount in a step-by-step manner. On the other hand, when the driving frequency f increases, the driving frequency f approaches the used frequency band FB1. Therefore, the first controller 25 may further increase the phase shift amount so that the driving frequency f is not included in the used frequency band FB1. Process similar to that of the above-described embodiment may be performed as process in a case where the frequency characteristic of the DC electric power P_{dc} has changed.

(86) In this modification, when the phase shift amount of the DC/AC converter 27 is changed, the frequency characteristic of the DC electric power P_{dc} is changed. Therefore, the driving frequency f is changed in order to maintain a state in which the power value of the DC electric power P_{dc} matches the first power command value. Thus, the driving frequency f can be set to a frequency different from the used frequency band. As described above, also in the power transmitter 2 of the modification, the same effects as those of the power transmitter 2 according to the embodiment are obtained.

(87) In the above modification, the first controller 25 controls the power converter 26 to keep the voltage V_{dc} constant. Specifically, the voltage V_{dc} is set to a low voltage such as the minimum voltage in the voltage range that can be output by the power converter 26. According to this configuration, the frequency change process can be performed without increasing the withstand voltage of the DC/AC converter 27. As a result, it is possible to suppress interference with other devices without increasing the size of the DC/AC converter 27. Since a component having a low withstand voltage can be used for the DC/AC converter 27, the cost of the power transmitter 2 can be reduced.

(88) When a switching element having a high withstand voltage is used as the switching element of the DC/AC converter 27, the on-resistance of the switching element increases, and thus the loss in the DC/AC converter 27 increases. Therefore, by using a switching element having a low withstand voltage, it is possible to reduce a decrease in the power efficiency of the power transmitter 2.

(89) Further, in the above modification, since the voltage V_{dc} is not changed, it is possible to follow a transient response such as a gap variation.

(90) The lower the voltage V_{dc} is, the higher the power transfer efficiency between the first coil 21 and the second coil 31 is. Therefore, the power transfer efficiency between the first coil 21 and the second coil 31 can be improved by lowering the voltage V_{dc} . By lowering the voltage V_{dc} , the possibility of breakdown of the power transmitter 2 can be reduced.

(91) Although the voltage V_{dc} can be further lowered by lowering the voltage V_{ac1} of the AC electric power P_{ac1} , the voltage range of the voltage V_{ac1} is narrowed. On the other hand, the power converter 26 can further lower the voltage V_{dc} without narrowing the voltage range of the voltage V_{ac1} by further including a step-down DC/DC converter in the subsequent stage of the PFC circuit. When the voltage V_{dc} is set within the voltage range that can be output from the PFC circuit, the step-down DC/DC converter can be omitted. Therefore, the cost of the power transmitter 2 can be reduced. In the above modification, the voltage V_{dc} may be changed by about

several volts.

(92) The first controller **25** may perform the frequency change process by performing the impedance control together with the frequency control. With reference to FIG. **10**, a wireless power transfer system **1** including a power transmitter **2** of another modification will be described. FIG. **10** is a circuit block diagram of a wireless power transfer system including a power transmitter according to another modification. As shown in FIG. **10**, the wireless power transfer system **1** according to another modification is mainly different from the wireless power transfer system **1** of the above-described embodiment in that the first converter **22** of the power transmitter **2** further includes an impedance converter **29**.

(93) The impedance converter **29** is provided between the DC/AC converter **27** and the first coil **21**. The impedance converter **29** is a device for changing the impedance between the DC/AC converter **27** and the first coil **21** (impedance viewed from the DC/AC converter **27**). The impedance converter **29** may be, for example, a tunable matching network (TMN). Since the TMN is known, a description thereof is omitted (refer to U.S. Patent Application Publication 2019/0006836, U.S. Patent Application Publication 2019/0006885, and the like).

(94) The first converter **22** can change the magnitudes (power values) of the DC electric power P_{dc} and the AC electric power P_{ac2} by the impedance control in addition to the frequency control, the phase shift control, and the voltage control of the DC electric power P_{dc} . That is, the power control performed by the first controller **25** is performed by using at least one of the frequency control, the phase shift control, the voltage control of the DC electric power P_{dc} , and the impedance control.

(95) Next, the impedance control will be described. The first controller **25** performs the impedance control of changing the magnitudes (power values) of the DC electric power P_{dc} , the AC electric power P_{ac2} , and the load electric power P_{out} by changing the magnitude of the impedance between the DC/AC converter **27** and the first coil **21**. The DC electric power P_{dc} and the AC electric power P_{ac2} decrease as the impedance between the DC/AC converter **27** and the first coil **21** increases, and the DC electric power P_{dc} and the AC electric power P_{ac2} increase as the impedance between the DC/AC converter **27** and the first coil **21** decreases. Therefore, the above-described power control parameter in the impedance control is the magnitude of the impedance between the DC/AC converter **27** and the first coil **21**.

(96) The power control of another modification differs from the power control of the above-described modification in that the impedance control is used instead of the phase shift control. Therefore, a detailed description thereof is omitted.

(97) In this modification, when the impedance between the DC/AC converter **27** and the first coil **21** is changed, the frequency characteristic of the DC electric power P_{dc} is changed. Therefore, the driving frequency f is changed in order to maintain a state in which the power value of the DC electric power P_{dc} matches the first power command value. Thus, the driving frequency f can be set to a frequency different from the used frequency band. As described above, also in the power transmitter **2** of another modification, the same effects as those of the power transmitter **2** according to the modification described above are obtained. The impedance converter **29** may be provided between the second coil **31** and the second converter **32**.

(98) The first controller **25** may select one of the voltage control of the DC electric power P_{dc} and the phase shift control in accordance with the driving frequency f (frequency f_0) when the power value of the DC electric power P_{dc} reaches the first power command value, and may perform the frequency change process by performing the selected control together with the frequency control. For example, it is assumed that the voltage V_{dc} is set to the minimum voltage in the voltage range that can be output by the power converter **26** and the phase shift amount is set to 0 before the frequency change process is performed. In this case, in order to maintain the state in which the power value of the DC electric power P_{dc} matches the first power command value, the driving frequency f can only be increased in the frequency change process by the power control of the DC electric power P_{dc} and the frequency control, and the driving frequency f can only be decreased in

the frequency change process by the phase-shift control and the frequency control.

(99) For example, the first controller **25** selects one of the power control of the DC electric power P_{dc} and the phase shift control of the DC/AC converter **27** in consideration of criteria such as whether the power value of the DC electric power P_{dc} can be caused to match the first power command value, whether the impedance of the entire load viewed from the DC/AC converter **27** (inverter circuit) becomes capacitive (C load) by the phase shift control, whether there is the influence of noise such as electromagnetic compatibility (EMC) and malfunction by the phase shift control, and whether the driving frequency f increases or decreases when the frequency characteristic of the DC electric power P_{dc} changes due to progress of power transfer. Here, the impedance of the entire load viewed from the DC/AC converter **27** (inverter circuit) in the present modification is the sum of the impedances of the impedance converter **29**, the first coil **21**, the second coil **31**, the second converter **32**, and the load L . For example, when the impedance of the entire load viewed from the DC/AC converter **27** (inverter circuit) is capacitive, the first controller **25** selects the voltage control of the DC electric power P_{dc} . When there is an influence of noise, the first controller **25** selects the voltage control of the DC electric power P_{dc} . Alternatively, the first controller **25** selects control in which the driving frequency f moves away from the used frequency band when the frequency characteristic of the DC electric power P_{dc} changes.

(100) This makes it possible to shorten the time until the driving frequency f is set to a frequency different from the used frequency band. Since the adjustment range of the driving frequency f can be widened by using the voltage control and the phase shift control, it is possible to more reliably set the driving frequency f to a frequency different from the used frequency band. Furthermore, it is possible to stabilize charging control while reducing the influence of noise without performing control for suppressing noise.

(101) Similarly, the first controller **25** may select one of the voltage control of the DC electric power P_{dc} and the impedance control in accordance with the driving frequency f (frequency f_0) when the power value of the DC electric power P_{dc} reaches the first power command value, and may perform the frequency change process by performing the selected control together with the frequency control. For example, it is assumed that before the frequency change process is performed, the voltage V_{dc} is set to the minimum voltage in the voltage range that can be output by the power converter **26**, and the impedance between the DC/AC converter **27** and the first coil **21** is set to the minimum. In this case, in order to maintain the state in which the power value of the DC electric power P_{dc} matches the first power command value, the driving frequency f can only be increased in the frequency change process by the power control and the frequency control, and the driving frequency f can only be decreased in the frequency change process by the impedance control and the frequency control.

(102) The criterion for selecting one of the voltage control of the DC electric power P_{dc} and the impedance control is the same as the criterion for selecting one of the voltage control of the DC electric power P_{dc} and the phase shift control. This makes it possible to shorten the time until the driving frequency f is set to a frequency different from the used frequency band. Since the adjustment range of the driving frequency f can be widened by using the voltage control and the impedance control, it is possible to more reliably set the driving frequency f to a frequency different from the used frequency band. Furthermore, it is possible to stabilize charging control while reducing the influence of noise without performing control for suppressing noise.

(103) As described above, the first controller **25** performs the frequency change process by performing at least one of the voltage control of the DC electric power P_{dc} , the phase shift control of the DC/AC converter **27**, and the impedance control between the DC/AC converter **27** and the first coil **21** together with the frequency control. When the voltage V_{dc} , the phase shift amount of the DC/AC converter **27**, or the impedance between the DC/AC converter **27** and the first coil **21** is changed, the frequency characteristic of the DC electric power P_{dc} is changed. Therefore, the driving frequency f is changed in order to maintain a state in which the power value of the DC

electric power P_{dc} matches the first power command value. Thus, the driving frequency f can be set to a frequency different from the used frequency band.

(104) As described above, the first controller **25** may control the first converter **22** so that the AC electric power P_{ac2} approaches the first power command value, which is the target value of the AC electric power P_{ac2} , as the constant power control. The AC electric power P_{ac2} has the same frequency characteristic as that of the DC electric power P_{dc} . The power control using the AC electric power P_{ac2} is the same as the power control using the DC electric power P_{dc} .

REFERENCE SIGNS LIST

(105) **1** wireless power transfer system **2** power transmitter **3** power receiver **4** first coil device **5** second coil device **21** first coil **22** first converter (converter) **23** first detector **24** first communication device **25** first controller (controller) **26** power converter **27** DC/AC converter **28** used frequency detector **29** impedance converter **31** second coil **32** second converter **33** second detector **34** second communication device **35** second controller SWa switching element SWb switching element SWc switching element SWd switching element EV electric vehicle FB1 used frequency band FB2 used frequency band I_{dc} electric current I_{out} load electric current L load P_{ac1} AC electric power P_{ac2} AC electric power P_{ac3} AC electric power P_{dc} DC electric power P_{out} load electric power PS power source R road surface Sa driving signal Sb driving signal Sc driving signal Sd driving signal V_{dc} voltage V_{out} load voltage

Claims

1. A power transmitter for supplying electric power to a power receiver connected to a load, the power transmitter comprising: a first coil configured to wirelessly transmit the electric power to a second coil of the power receiver; a converter including a DC/AC converter configured to convert DC electric power into AC electric power and supply the AC electric power to the first coil; and a controller configured to bring a power value of target electric power close to a power command value by frequency control of changing a frequency of the AC electric power, wherein the target electric power is the DC electric power or the AC electric power, wherein the controller performs, when the frequency at which the power value reaches the power command value is included in a used frequency band used by another device, frequency change process of changing the frequency so that the frequency becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value matches the power command value, and wherein the controller: selects one of voltage control of changing a voltage of the DC electric power and phase shift control of the DC/AC converter in accordance with the frequency when the power value reaches the power command value, and performs the frequency change process by performing the selected control together with the frequency control.
2. The power transmitter according to claim 1, wherein the controller performs the voltage control by increasing the voltage when the voltage control is performed together with the frequency control.
3. The power transmitter according to claim 1, wherein the controller controls the converter to keep the voltage constant when the phase shift control is performed together with the frequency control.
4. The power transmitter according to claim 1, wherein the controller decreases the power command value when the controller cannot change the frequency to any frequency different from the used frequency band.
5. A power transmitter for supplying electric power to a power receiver connected to a load, the power transmitter comprising: a first coil configured to wirelessly transmit the electric power to a second coil of the power receiver; a converter including a DC/AC converter configured to convert DC electric power into AC electric power and supply the AC electric power to the first coil; and a controller configured to bring a power value of target electric power close to a power command value by frequency control of changing a frequency of the AC electric power, wherein the target

electric power is the DC electric power or the AC electric power, wherein the controller performs, when the frequency at which the power value reaches the power command value is included in a used frequency band used by another device, frequency change process of changing the frequency so that the frequency becomes a frequency different from the used frequency band while performing constant power control of maintaining a state in which the power value matches the power command value, and wherein the controller: selects one of voltage control of changing a voltage of the DC electric power and impedance control of controlling an impedance between the DC/AC converter and the first coil in accordance with the frequency when the power value reaches the power command value, and performs the frequency change process by performing the selected control together with the frequency control.

6. The power transmitter according to claim 5, wherein the controller controls the converter to keep the voltage constant when the impedance control is performed together with the frequency control.
